



Submitted by

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Semester: 4th

Section: BS-SE (Morning)

Assignment: 1st

Subject: Database Systems

Submitted to

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Instructions

For each question, you must perform the following tasks:

- a) List out entity classes along with appropriate attributes.
- b) Identify all necessary relationships.
- c) Identify the connectivity for each relationship.
- d) Draw ER / EER model using an appropriate CASE tool, such as MS Visio, Lucid chart etc.
- e) You can make your assumptions (if needed).

question Set – 1 [4 Questions, 10 Marks Each]

➤ Question#1:

In the PDO Company, we have identified the following entity type. An identifier is also suggested for each entity, together with selected important attributes. The company receives many orders from different customers. The identifier for an Employee is EMPLOYED the following additional attributes are Employee Name, Employee Number, Employee birth date, Age which will get based on the Birthday and Employee qualification (some employees are having more than One qualification and we have other Entities In these company Sales, STOCK and INVENTORY for Employees who are working here and The FUEL entity have Flowing attributes are Price of Fuel and Number of Tanks. An employee delivers any number of orders, and each order must belong to exactly one employee. custom MER Submit order. The identity of ORDER is orderNumber. Other attributes include order-Name and order-Date.

Services:

- Customer can place order
- Order taking Employee will receive customer request.
- Employee will check the inventory and proceed the customer request.
- Employees have their designations according to their qualifications in the company.
- Delivery employee receive the fuel tanks from company to deliver the order to customer

➤ Answer#1:

➤ Task#1: Entity Classes & Attributes

1. Employee
 - *Attributes:* Employee-ID, Name, Birth-Date, Age (derived), Qualification
2. Customer
 - *Attributes:* Customer-ID, Name, Contact-Info (optional), Address
3. Order

- *Attributes:* Order-Number, Order-Name, Order-Date
- 4. Qualification
 - *Attributes:* Qualification-ID, Qualification-Name
- 5. Inventory
 - *Attributes:* Item-Name, Quantity
- 6. Fuel
 - *Attributes:* Price, Number-of-Tanks
- 7. Sales
 - *Attributes:* Sale-ID, Sale-Date, Total-Amount
- 8. Stock
 - *Attributes:* Stock-ID, Stock-Name, Quantity

➤ Task#2: Relationships

1. Has
 - Each employee has at least one qualification → total participation from the Employee
 - Use associative entity: EmployeeQualification
2. Process
 - Each order is processed by one employee; one employee can process multiple orders
3. Place
 - A customer may place multiple orders
4. O_I
 - Orders require items from inventory; inventory items can be used in multiple orders
5. O_F
 - Each order may include several fuel purchases (Fuel entries); each fuel entry belongs to one order
6. O_S
 - One order leads to a sale; can be 1:M if partial deliveries are tracked
7. F_S
 - Fuel usage affects stock levels (fuel entry reduces stock)

➤ Task#3: Connectivity Summary

Relationship	Connectivity
Has	1:M
Process	1:1 or M:0
place	1:M or 1:1

Relationship

O_I

O_F

O_S

F_S

Connectivity

M:1 or M:0

1:M

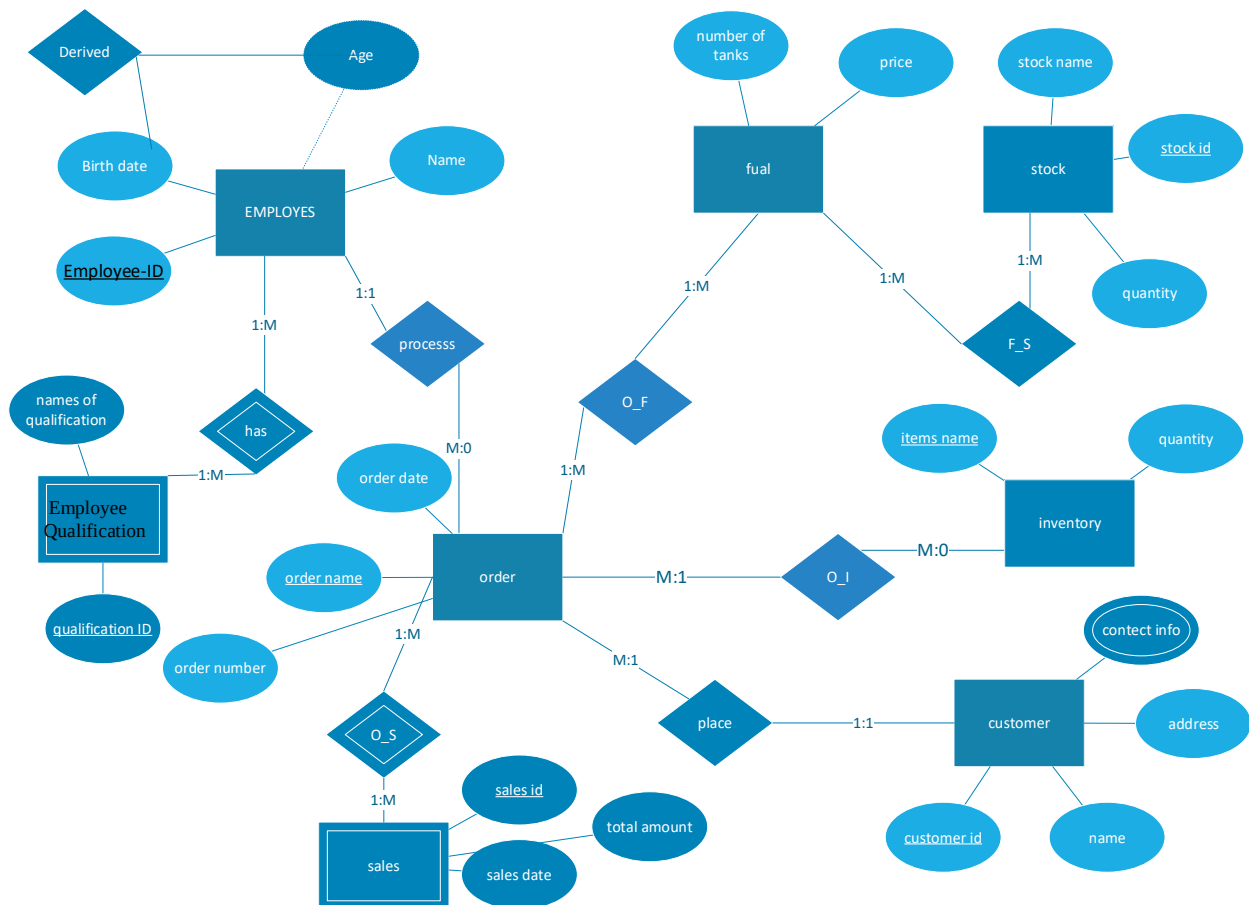
1:1 or 1:M

1:M

➤ Task#4 Assumptions

1. Age is a derived attribute of Employee (from Birthdate).
2. Use a bridge entity, Employee Qualification, to resolve the M:N total relationship.
3. Fuel is connected only to Order and Stock, not to Employee.
4. Stock may be updated based on Fuel usage, especially for available tanks

➤ Task#5 ER Diagram



Question#2:

Prepare an ERD for a real estate firm that lists property for sale. Also prepare a definition for each entity type, attribute, and relationship on your diagram. The following describes this organization:

- The firm has a number of sales offices in several states. Attributes of sales office include Office Number (identifier) and Location.
- Each sales office is assigned one or more employees. Attributes of employee include Employee ID (identifier) and Employee Name. An employee must be assigned to only one sales office.
- For each sales office, there is always one employee assigned to manage that office. An employee may manage only the sales office to which he or she is assigned. • The firm lists property for sale. Attributes of property include Property ID (identifier) and Location. Components of Location include Address, City, State, and Zip Code.
- Each unit of property must be listed with one (and only one) of the sales offices. A sales office may have any number of properties listed or may have no properties listed.
- Each unit of property has one or more owners. Attributes of owners are Owner ID (identifier) and Owner Name. An owner may own one or more units of property. An attribute of the relationship between property and owner is Percent Owned.

➤ Answer#2:

➤ Task#1: Entity Classes & Attributes

1. SalesOffice
 - *Attributes:* OfficeNumber, Location
2. Employee
 - *Attributes:* EmployeeID, EmployeeName, officeNumber
3. Property
 - *Attributes:* PropertyID, officeNumber
4. Owner
 - *Attributes:* OwnerID, OwnerName
5. Location
 - *Attribute:* Address, City, State, ZipCode

➤ Task#2: Relationships

1. Assigned To
 - *Between:* Employee and Sales Office
 - *Description:* Each employee is assigned to exactly one sales office.
2. Manager
 - *Between:* Employee and Sales Office
 - *Description:* Each office has one manager who is also assigned to that office.
3. Listed At
 - *Between:* Property and Sales Office
 - *Description:* Each property is listed at exactly one sales office.
4. Owns
 - *Between:* Owner and Property

- *Description:* Each property has one or more owners, and each owner may own multiple properties.
 - *Attribute of relationship:* Percent Owned
5. *Includes*
- *Between:* Property and Location
 - *Description:* Each property includes a location which has components like address, zip code etc.
 - *Attribute of Relationship:* Property contained

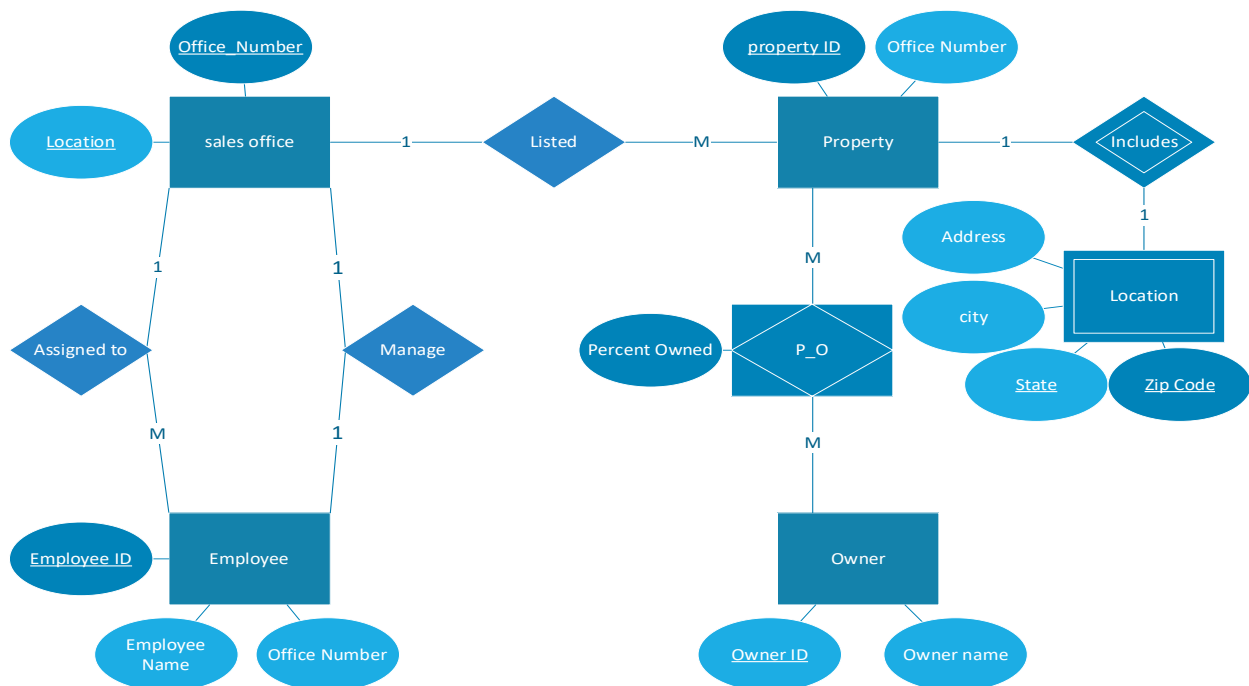
➤ Task#3: Connectivity Summary

Relationship	Connectivity
Assigned To	1:N (One Sales Office → Many Employees)
Manages	1:1 (Each Sales Office has one Manager who is an Employee assigned to it)
Listed At	0:M (One Sales Office → Many Properties or no properties) 1:1 (Each unit of property → only one sale office)
Includes	1:1 (Each property → Each location)
Owns	M:N (Each Owner → Many Properties, Each Property → Many Owners)

➤ Task#4 Assumption

1. Adding the office_number attribute to the employee and property entity,
2. Assuming location as a separate entity, it is composed of the attributes, and considering zip code as the location identifier.

➤ Task#5 ER Diagram



❖ Definition

➤ ENTITY TYPES

1. Sales Office
 - Represents a real estate sales office responsible for listing properties and assigning employees.
2. Employee
 - Represents individuals working in the sales offices. Each employee may be assigned to and manage a sales office.
3. Property
 - Represents real estate units being sold or rented. Properties are listed by sales offices and owned by individuals/entities.
4. Location
 - Represents the physical location details of properties, including address, city, state, and zip code.
5. Owner
 - Represents individuals or entities that have ownership (full or partial) of properties.

➤ Attributes

1. Sales Office
 - Office_Number: Unique identifier for each sales office.
 - Location: Describes the physical placement of the office.
2. Employee Attributes
 - Employee_ID: Unique identifier for each employee.
 - Employee_Name: Full name of the employee.
 - Office_Number: The office number where the employee works.
3. Property Attributes
 - Property_ID: Unique identifier for each property.
 - Office Number: Identifies which office listed the property.
4. Location Attributes (for Property Location)
 - Address: Street address of the property.
 - City: City where the property is located.
 - State: State where the property is located.
 - Zip Code: Zip/postal code of the property's location.
5. Owner Attributes
 - Owner_ID: Unique identifier for each owner.
 - Owner_Name: Full name of the owner.
6. Relationship Attribute
 - Percent Owned (in P_O relationship): Percentage of ownership an owner holds in a specific property.

➤ RELATIONSHIPS

1. Assigned To
 - Between: Employee and Sales Office
 - Cardinality: Many employees (1:M) can be assigned to one sales office
 - Description: Indicates which office an employee is assigned to.
2. Manage
 - Between: Employee and Sales Office
 - Cardinality: One-to-one (1:1)
 - Description: Shows which employee manages a specific sales office.
3. Listed
 - Between: Sales Office and Property
 - Cardinality: One-to-many (1:M)
 - Description: Specifies which properties are listed by a sales office.
4. Includes
 - Between: Property and Location
 - Cardinality: One-to-one (1:1)
 - Description: Links each property to its unique location.
5. P_O (Property–Owner)
 - Between: Property and Owner
 - Cardinality: Many-to-many (M:N)
 - Attribute: Percent Owned
 - Description: Indicates ownership relationships between owners and properties, including the ownership percentage.

Question#3:

Create an ERD for a car dealership. The dealership sells both new and used cars, and it operates a service facility. Base your design on the following business rules:

- A salesperson may sell many cars, but each car is sold by only one salesperson.
- A customer may buy many cars, but each car is bought by only one customer.
- A salesperson writes a single invoice for each car he or she sells.
- A customer gets an invoice for each car he or she buys.
- A customer may come in just to have his or her car serviced; that is, a customer need not buy a car to be classified as a customer.
- When a customer takes one or more cars in for repair or service, one service ticket is written for each car.
- The car dealership maintains a service history for each of the cars serviced. The service records are referenced by the car's serial number.
- A car brought in for service can be worked on by many mechanics, and each mechanic may work on many cars.

- A car that is serviced may or may not need parts (e.g., adjusting a carburetor or cleaning a fuel injector nozzle does not require providing new parts).

➤ Answer#3:

➤ Task#1: Entity Classes & Attributes

1. Salesperson
 - Attributes: SalespersonID, Name, Contact Info
2. Customer
 - Attributes: CustomerID, Name, Contact Info, Address
3. Car
 - Attributes: SerialNumber, Model, Year, Type (New/Used)
4. Invoice
 - Attributes: InvoiceID, Date, Amount, CarSerialNumber (FK), SalespersonID (FK), CustomerID (FK)
5. ServiceTicket/Repair
 - Attributes: TicketID, Date, Description, CarSerialNumber (FK), CustomerID (FK)
6. Mechanic
 - Attributes: MechanicID, Name, Specialty
7. ServiceHistory
 - Attributes: EntryID, CarSerialNumber (FK), ServiceDate, Description, PartsUsed (Optional)

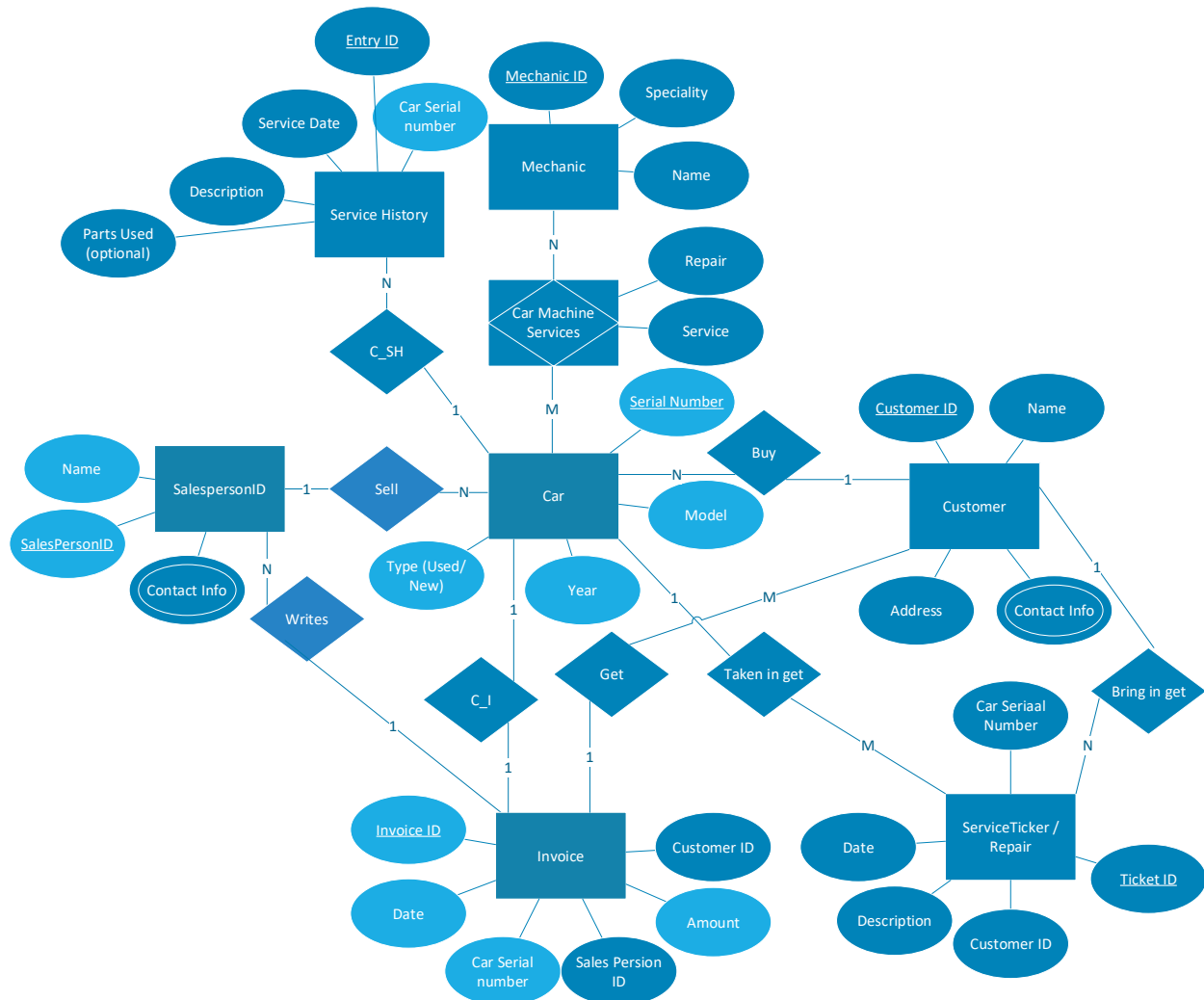
➤ Task#2: Relationships

1. Sell
 - One-to-many (A salesperson sells many cars, each car sold by only one salesperson)
2. Buy
 - One-to-many (Customer may buy many cars, each car sold to only one customer)
3. Writes
 - One-to-many (One invoice for car, per customer)
4. Gets
 - One-to-many (One invoice per car, per customer)
5. Car–Invoice
 - One-to-one (Each car has exactly one invoice)
6. Bring in get
 - One-to-many (Customer may bring many cars for service)
7. Taken in get
 - One-to-many (Each car may have many service tickets)
8. Car–Mechanic (via service)
 - Many-to-many (through a CarMechanicService associative entity)
9. Car–ServiceHistory
 - One-to-many (Each car can have multiple service records)

➤ Task#3: Connectivity Summary

Relationship	Connectivity
Sell	1:N
Buy	1:N
Writes	1:N
Get	1:N
Car – Invoice	1:1
Bring in get	1:N
Taken in get	1:N
Mechanic – Car (Service)	M:N (via junction)
Car – ServiceHistory	1:N

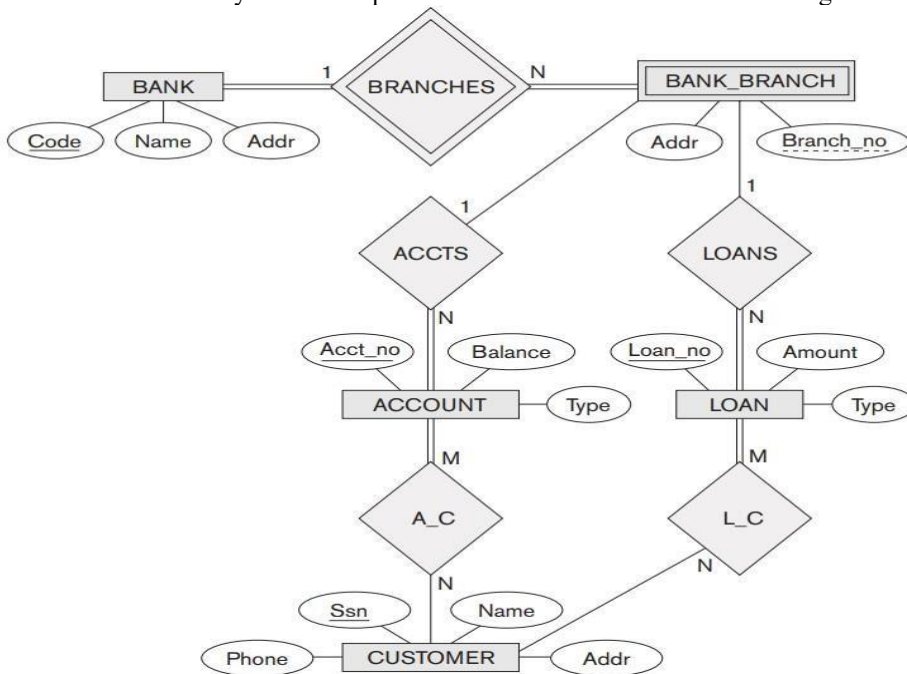
➤ Task#4 ER Diagram



Question#4:

Consider the ER diagram shown in the following figure for part of a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.

- List the strong entity types in the ER diagram.
- Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
- What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?
- List the names of all relationships.
- List concisely the user requirements that led to this ER schema design.



➤ Answer#4:

➤ Task#1: Entity Classes & Attributes

1. BANK
 - Attributes: Code, Name, Addr
2. BANK_BRANCH
 - Attributes: Branch_no, Addr
3. ACCOUNT
 - Attributes: Acct_no, Balance, Type
4. LOAN
 - Attributes: Loan_no, Amount, Type
5. CUSTOMER
 - Attributes: Ssn, Name, Phone, Addr

➤ Task#2: Relationships

1. BRANCHES
 - Between: BANK (1) and BANK_BRANCH (N)
 - Meaning: One bank has many branches.

2. ACCTS

- Between: BANK_BRANCH (1) and ACCOUNT (N)
- Meaning: Each branch maintains multiple accounts.

3. LOANS

- Between: BANK_BRANCH (1) and LOAN (N)
- Meaning: Each branch issues multiple loans.

4. A_C (Account-Customer)

- Between: ACCOUNT (M) and CUSTOMER (N)
- Meaning: Many customers can own many accounts (joint accounts possible).

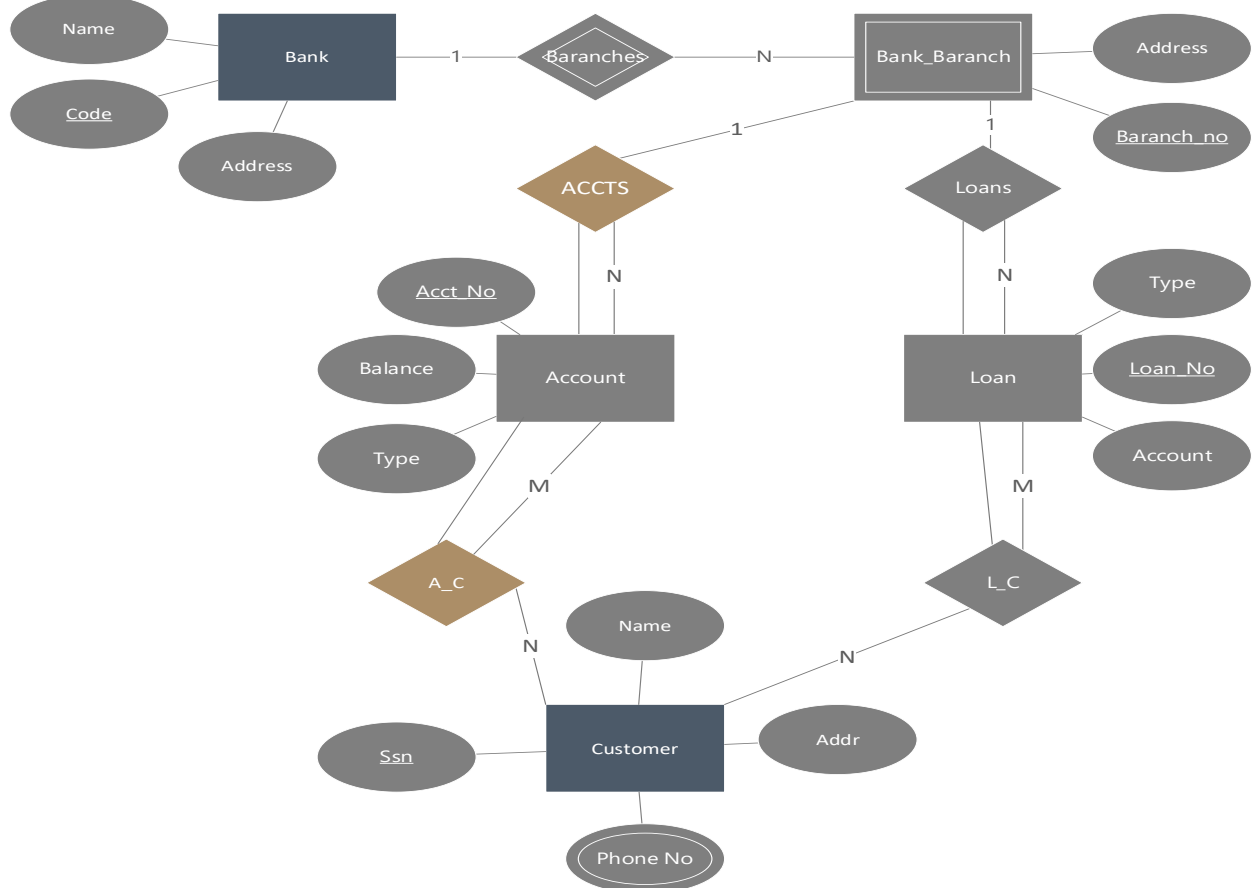
5. L_C (Loan-Customer)

- Between: LOAN (M) and CUSTOMER (N)
- Meaning: Many customers can share many loans.

➤ Task#3: Connectivity Summary

Relationship	Connectivity
BRANCHES	1:N
ACCTS	1:N
LOANS	1:N
A_C	M:N
L_C	M:N

➤ Task#4 ER Diagram



❖Detail

➤ List of Strong Entity Types

1. BANK – identified by Code
2. ACCOUNT – identified by Acct_no
3. LOAN – identified by Loan_no
4. CUSTOMER – identified by Ssn

➤ Is there any Weak Entity Type

Yes – BANK_BRANCH appears to be a weak entity:

- Weak Entity Type (Name): BANK_BRANCH
- Partial Key: Branch_no
- Identifying Relationship: BRANCHES (with BANK, 1:N)

➤ Constraints on the Weak Entity

The weak entity BANK_BRANCH:

- Depends on the BANK for identification.
- Has a partial key, Branch_no.
- The identifying relationship BRANCHES is total participation from the weak entity (BANK_BRANCH) and has a 1:N cardinality from BANK to BANK_BRANCH.

So, constraints:

- Each BANK_BRANCH must be related to exactly one BANK.
- A BANK can have many BANK_BRANCHES.
- The combination (Code+Branch_no) uniquely identifies each BANK_BRANCH.

➤ Names of All Relationships

1. BRANCHES – between BANK and BANK_BRANCH
2. ACCTS – between BANK_BRANCH and ACCOUNT
3. LOANS – between BANK_BRANCH and LOAN
4. A_C – between ACCOUNT and CUSTOMER (many-to-many)
5. L_C – between LOAN and CUSTOMER (many-to-many)

➤ User Requirements Behind the ER Schema Design:

1. A bank operates multiple branches, each uniquely identified by a branch number within its parent bank.
2. Each branch:
 - Has a physical address.
 - Maintains multiple customer accounts and issues loans.
3. Accounts:
 - Are uniquely identified by an account number.
 - Have associated details like balance and type.
 - Are linked to the branch where they are held.
4. Loans:
 - Are uniquely identified by a loan number.
 - Include amount and type.
 - Are also issued from a specific branch.
5. Customers:
 - Can own multiple accounts and take multiple loans.

- May share accounts or loans with other customers (many-to-many relationships).
 - Are uniquely identified by their SSN, and also store name, address, and phone.
6. Data integrity:
- Branches must belong to a bank.
 - Accounts and loans must belong to a branch.
 - The schema must track customer relationships to both accounts and loans accurately.
-

question Set – 2 [4 Questions, 15 Marks Each]

Question# 1:

During peak periods, Temporary Employment Corporation (TEC) places temporary workers in companies.

TEC's manager gives you the following description of the business:

- TEC has a file of candidates who are willing to work.
- Any candidate who has worked before has a specific job history. (Naturally, no job history exists if the candidate has never worked.) Each time the candidate works, one additional job history record is created.
- Each candidate has earned several qualifications. Each qualification may be earned by more than one candidate. (For example, more than one candidate may have earned a Bachelor of Business Administration degree or a Microsoft Network Certification, and clearly a candidate may have earned both a BBA and a Microsoft Network Certification.)
- TEC offers courses to help candidates improve their qualifications.
- Every course develops one specific qualification; however, TEC does not offer a course for every qualification. Some qualifications are developed through multiple courses.
- Some courses cover advanced topics that require specific qualifications as prerequisites. Some courses cover basic topics that do not require any prerequisite qualifications. A course can have several prerequisites. A qualification can be a prerequisite for more than one course.
- Courses are taught during training sessions. A training session is the presentation of a single course. Over time, TEC will offer many training sessions for each course; however, new courses may not have any training sessions scheduled right away.
- Candidates can pay a fee to attend a training session. A training session can accommodate several candidates, although new training sessions will not have any candidates registered at first.
- TEC also has a list of companies that request temporaries.
- Each time a company requests a temporary employee, TEC makes an entry in the Openings folder. That folder contains an opening number, a company name, required qualifications, a starting date, an anticipated ending date, and hourly pay.
- Each opening requires only one specific or main qualification.
- When a candidate matches the qualification, the job is assigned, and an entry is made in the Placement Record folder. The folder contains such information as an opening number,

candidate number, and total hours worked. In addition, an entry is made in the job history for the candidate.

- An opening can be filled by many candidates, and a candidate can fill many openings.
- TEC uses special codes to describe a candidate's qualifications for an opening. The list of Codes are:

CODE	DESCRIPTION
SEC-45	Secretarial work; candidate must type at least 45 words per minute
SEC-60	Secretarial work; candidate must type at least 60 words per minute
CLERK	General clerking work
PRG-VB	Programmer, Visual Basic
PRG-C++	Programmer, C++
DBA-ORA	Database Administrator, Oracle
DBA-DB2	Database Administrator, IBM DB2
DBA-SQLSERV	Database Administrator, MS SQL Server
SYS-1	Systems Analyst, level 1
SYS-2	Systems Analyst, level 2
NW-NOV	Network Administrator, Novell experience
WD-CF	Web Developer, ColdFusion

➤ Answer#1:

➤ Task#1: Entity Classes & Attributes

1. Candidate
 - Attributes: CandidateID, Name, Contact Info, Total Experience (derived), Address
2. . Job History
 - Attributes: JobHistoryID, CandidateID, JobTitle, StartDate, EndDate, CompanyName
3. Qualification
 - Attributes: QualificationID, QualificationName, Description
4. Course
 - Attributes: CourseID, CourseName, Fee
5. Prerequisite
 - Attributes: QualificationID, CourseID
6. Training Session
 - Attributes: SessionID, CourseID, Date, Location
7. Company
 - Attributes: CompanyID, CompanyName, Contact Info
8. Opening
 - Attributes: OpeningID, CompanyID, QualificationID, StartDate, EndDate, HourlyPay
9. Placement
 - Attributes: PlacementID, OpeningID, CandidateID, HoursWorked
10. Code
 - Attributes: CodeID, Description

➤ Task#2: Relationships

1. Candidate – Qualification
 - Each candidate can have many qualifications, and each qualification can be held by many candidates → M:N
2. Course – Qualification (Develops)

- Many course develops one qualification, a qualification can be developed by many courses → M:N
- 3. Course – Prerequisite Qualification
 - Each course may have multiple prerequisite qualifications → M:N
- 4. Course – Training Session
 - Each course can have multiple training sessions → 1:M
- 5. Candidate – Training Session
 - A candidate can attend many training sessions → M:N
- 6. Candidate – Job History
 - A candidate may have multiple job history records → 1:M
- 7. Company – Opening
 - A company may post many job openings → 1:M
- 8. Opening – Qualification
 - Each opening requires one main qualification → M:1
- 9. Company – Code
 - A company has many special code, Many special codes are used by a company → 1:N via Placement
- 10. Candidate – Opening – Placement
 - A candidate can fill many openings, and an opening can be filled by many candidates → M:N via Placement

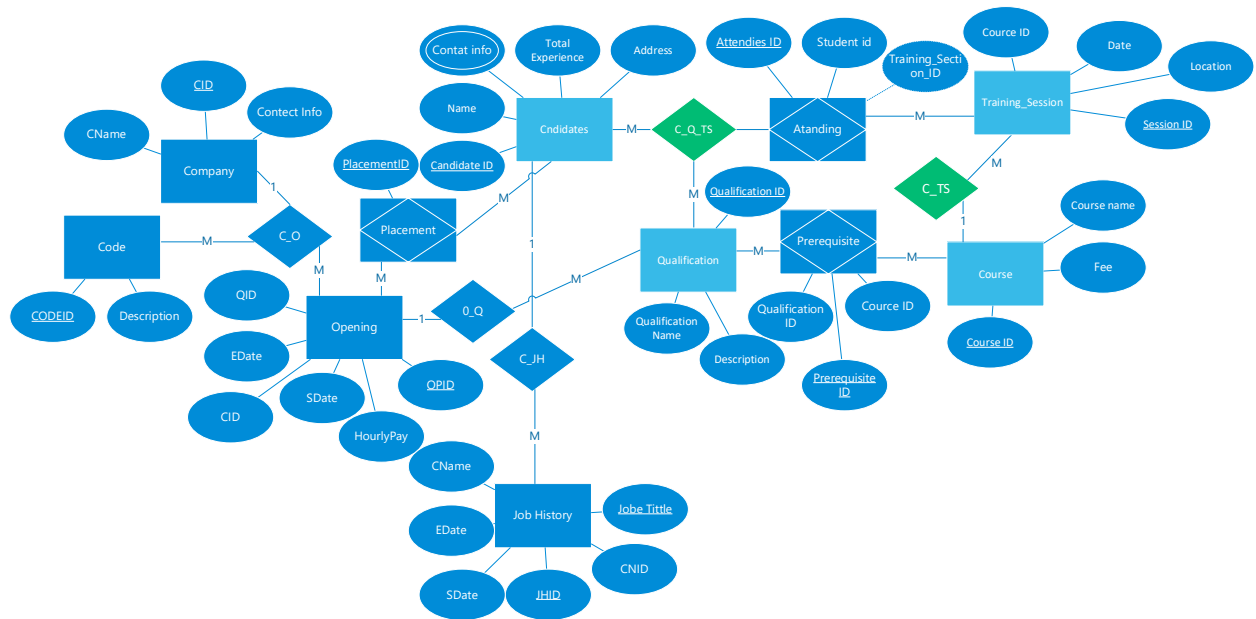
➤ Task#3: Connectivity Summary

Relationship	Connectivity
Candidate – Qualification	M:N
Company -- Code	1:M
Course – Qualification (develops)	M:N
Course – Prerequisite Qualification	M:N
Course – Training Session	1:M
Candidate – Training Session	M:N
Candidate – Job History	1:M
Company – Opening	1:M
Opening – Qualification	M:1
Candidate – Opening (Placement)	M:N (via Placement)

➤ Task#4 Assumption

1. Total Experience is a derived attribute of Candidate (from Job History).
10. Use a bridge entity for M:N relationships such as CandidateQualification and CoursePrerequisite.
10. Placement Record serves as an associative entity between Candidate and Opening with additional attribute (HoursWorked).
10. Each course develops one qualification, but qualifications may be developed by many courses.

➤ Task#5 ER Diagram



Question# 2:

An international school of technology has hired you to create a database management system to assist in scheduling classes. After several interviews with the president, you have come up with the following list of entities, attributes, and initial business rules:

- Room is identified by Building ID and Room No and also has a Capacity. A room can be either a lab or a classroom. If it is a classroom, it has an additional attribute called Board Type.
 - Media is identified by MType ID and has attributes of Media Type and Type Description. Note: Here we are tracking type of media (such as a VCR, projector, etc.), not the individual piece of equipment. Tracking of equipment is outside the scope of this project.
 - Computer is identified by CType ID and has attributes Computer Type, Type Description, Disk Capacity, and Processor Speed. Please note: As with Media Type, we are tracking only the type of computer, not an individual computer. You can think of this as a class of computers (e.g., PIII 900MHZ).
 - Instructor has identifier Emp ID and has attributes Name, Rank, and Office Phone.
 - Timeslot has identifier TSIS and has attributes Day Of Week, Start Time, and End Time.
 - Course has identifier Course ID and has attributes Course Description and Credits. Courses can have one, none, or many prerequisites. Courses also have one or more sections.
 - Section has identifier Section ID and attribute Enrollment Limit. After some further discussions, you have come up with some additional business rules to help you create the initial design:
 - An instructor teaches one, none, or many sections of a course in a given semester.
 - An instructor specifies preferred time slots.
 - Scheduling data are kept for each semester, uniquely identified by semester and year.
 - A room can be scheduled for one section or no section during one time slot in a given semester of a given year. However, one room can participate in many schedules, one schedule, or no schedules; one time slot can participate in many schedules, one schedule, or no schedules; one section can participate in many schedules, one schedule, or no schedules. Hint: Can you associate this to anything that you have seen before?
 - A room can have one type of media, several types of media, or no media.
 - Instructors are trained to use one, none, or many types of media.
 - A lab has one or more computer types. However, a classroom does not have any computers.
- A room cannot be both a classroom and a lab. There also are no other room types to be incorporated into the system.

➤ Answer#2:

➤ Task#1: Entity Classes & Attributes

1. Room
 - Attributes: BuildingID, RoomNo , Capacity
2. Media
 - Attributes: MTypeID, MediaType, TypeDescription
3. Computer
 - Attributes: CTypeID, ComputerType, TypeDescription, DiskCapacity, ProcessorSpeed
4. Instructor
 - Attributes: EmpID, Name, Rank, OfficePhoneNumber
5. Timeslot
 - Attributes: TSID, DayOfWeek, StartTime, EndTime
6. Course
 - Attributes: CourseID, CourseDescription, Credits
7. Section
 - Attributes: SectionID, EnrollmentLimit, CourseID
8. Semester
 - Attributes: SemesterID, Year
9. Schedule
 - Attributes: ScheduleID (*or composite PK: SectionID, Semester, Year, TSID*), SectionID, TSID, BuildingID + RoomNo, Semester, Year
10. Lab (Type of class room)
 - Attributes: LabID
11. Classroom (Type of room)
 - Attributes: ClassID, BoardType

➤ Task#2: Relationships

1. Types(Have)
 - A room can have many media types, and each media type can exist in many rooms → M:N
2. Types(Use)
 - An instructor can use many media types, and each media type can be used by many instructors → M:N
3. Lab (RoomType = 'Lab') – Computer (LabComputer)
 - A lab can have multiple types of computers → 1:M
4. Instructor – Section (Teaches)
 - An instructor can teach multiple sections → 1:M
5. Course – Course (Prerequisite)
 - A course may have many prerequisites, and a course can be a prerequisite for many other courses → M:N (Recursive)
6. Instructor – Timeslot (Prefers)
 - An instructor can prefer multiple timeslots, and a timeslot can be preferred by multiple instructors → M:N

7. Section – Room/Timeslot/Semester (Schedule)
 - A section is scheduled in a specific room, timeslot, and semester → M:1 via Schedule
8. Course – Section
 - A course can be introduced in multiple sections. Multiple sections can provide the same course → 1:M

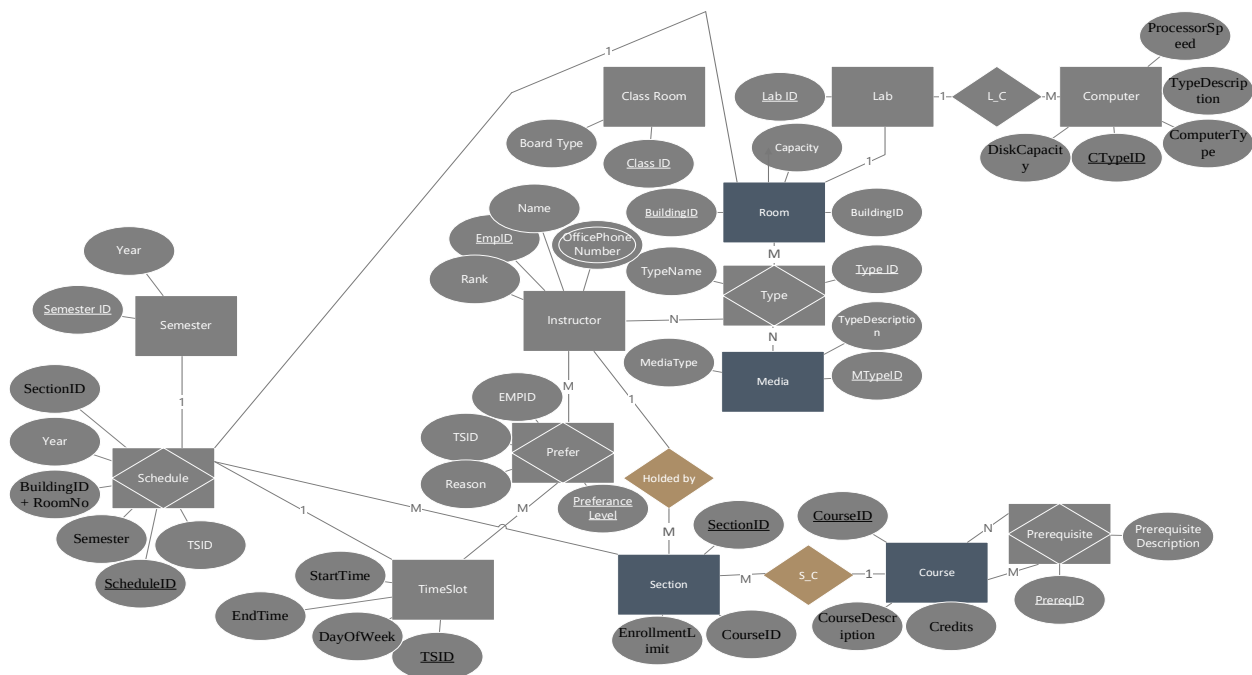
➤ Task#3: Connectivity Summary

Relationship	Connectivity
Types(Have)	M:N
Types(Use)	M:N
Lab — ComputerType	1:M
Instructor — Section	1:M
Course — Prerequisite Course	M:N (recursive)
Course -- Section	1:M
Instructor — Timeslot (Preference)	M:N
Section — Schedule (Room, Timeslot, Semester) M:1	

➤ Task#4 Assumption

1. A lab is a type of room with at least one computer type; a classroom cannot have computers.
2. The RoomType attribute helps define whether board type or computers apply.
3. Scheduling is unique per (Room, Timeslot, Semester, Year), ensuring no double-bookings.

➤ Task#5 ER Diagram



Question#3:

A healthcare management system is designed to manage patient information, appointments, medical records, healthcare professionals, and billing processes. The system has the following modules:

- **Patient Registration:**
A Patient registers with the hospital, providing personal information such as name, date of birth, contact details, and insurance information. Each Patient can have multiple visits to the hospital.
- **Appointment Scheduling:**
Patients can schedule Appointments with Healthcare Professionals (Doctors, Nurses). Each Appointment includes a date, time, and purpose of the visit. A Healthcare Professional can have multiple Appointments, and each Appointment can involve multiple Patients in group settings.
- **Healthcare Professionals:**
Healthcare Professionals have various specializations (e.g., General Practitioner, Cardiologist, Surgeon) and are affiliated with specific Departments within the hospital. Each Department can have multiple Healthcare Professionals.
- **Medical Records:**
Each Patient has a Medical Record that includes details of their medical history, medications, allergies, and treatment plans. Each Medical Record can have multiple entries over time, and a Healthcare Professional is responsible for updating it during each visit.
- **Billing and Insurance:**
Each Appointment generates a Bill, which includes charges for services rendered, medications prescribed, and any insurance claims submitted. Patients can have multiple Bills, and each Bill can be associated with one or more Insurance Claims.
- **Pharmacy Management:**
The hospital has a Pharmacy that manages medication inventory. Healthcare Professionals can prescribe medications, which are linked to specific Patients and included in their Medical Records.
- **Lab Tests and Results:**
Patients may undergo Lab Tests (e.g., blood tests, X-rays), which are ordered by Healthcare Professionals.
Each Lab Test has associated Results that are recorded in the Patient's Medical Record.
- **Emergency Services:**
The system includes Emergency Services where Patients can be admitted directly. Emergency Cases are recorded separately and can result in immediate treatment plans and urgent billing procedures.

➤ Answer#3:

➤ Task#1: Entity Classes & Attributes

- **Patient**
 - Attributes: Patient-ID, Name, Date-of-Birth, Contact-Info, Insurance-Info
- **HealthcareProfessional**
 - Attributes: HP-ID, Name, Specialization, Department-ID
- **Department**
 - Attributes: Department-ID, Department-Name

- **Appointment**
 - Attributes: Appointment-ID, Date, Time, Purpose
- **MedicalRecord**
 - Attributes: Record-ID, Created-Date
- **MedicalEntry**
 - Attributes: Entry-ID, Visit-Date, Details, Medications, Allergies, Treatment-Plans
- **Bill**
 - Attributes: Bill-ID, Appointment-ID, Total-Charges
- **InsuranceClaim**
 - Attributes: Claim-ID, Amount-Claimed, Status
- **Medication**
 - Attributes: Medication-ID, Name, Dosage
- **Prescription**
 - Attributes: Prescription-ID, Date, Notes
- **LabTest**
 - Attributes: LabTest-ID, Type, Ordered-Date
- **LabResult**
 - Attributes: Result-ID, LabTest-ID, Result-Date, Findings
- **EmergencyCase**
 - Attributes: Case-ID, Arrival-Date, Treatment-Plan, Urgency-Level
- **Pharmacy**
 - Attributes: Pharmacy-ID, Location

➤ Task#2: Relationships

1. **Registers**
 - A Patient registers multiple Appointment
→ 1:M from Patient to Appointment
2. **Schedules**
 - A Patient schedules many Appointments
 - Each Appointment may involve multiple Patients (for group settings)
→ M:N between Patient and Appointment
3. **AssignedTo**
 - Each Appointment is with one or more Healthcare Professionals
→ M:N between Appointment and HealthcareProfessional
4. **BelongsTo**
 - Each Healthcare Professional belongs to one Department
→ M:1 from HealthcareProfessional to Department
5. **Has**
 - Each Patient has one MedicalRecord
 - A MedicalRecord has many MedicalEntries
→ 1:1 from Patient to MedicalRecord, 1:M from MedicalRecord to MedicalEntry
6. **Updates**
 - A Healthcare Professional updates a Patient's MedicalRecord
→ M:N between MedicalEntry and HealthcareProfessional

7. **Generates**
 - Each Appointment generates one Bill
→ 1:1 from Appointment to Bill
8. **Has**
 - A Bill may have multiple Insurance Claims
→ 1:M from Bill to InsuranceClaim
9. **Prescribes**
 - A Healthcare Professional prescribes Medications to a Patient
→ M:N via Prescription entity (Prescription ↔ Medication ↔ Patient)
10. **ManagesInventory**
 - Pharmacy manages Medication inventory
→ M:N between Pharmacy and Medication
11. **OrdersTest**
 - A Healthcare Professional orders Lab Tests
→ 1:M from HealthcareProfessional to LabTest
12. **HasResults**
 - Each Lab Test has one or more Lab Results
→ 1:M from LabTest to LabResult
13. **InvolvedIn**
 - Patients can be admitted through Emergency Services
→ 1:M from Patient to EmergencyCase
14. **Treated**
 - A Healthcare Professional may treat Emergency Cases
→ M:N between HealthcareProfessional and EmergencyCase

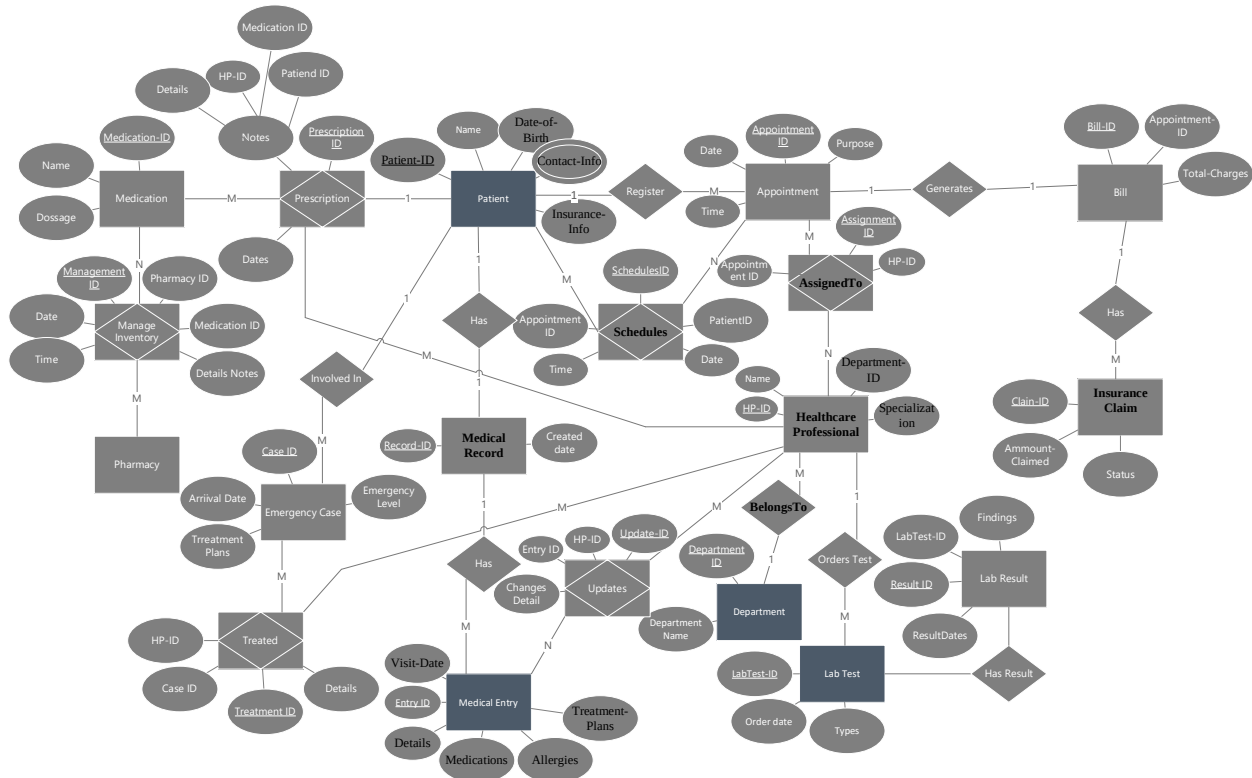
➤ Task#3: Connectivity Summary

Relationship	Connectivity
Registers / Schedules	M:N (Patient ↔ Appointment)
AssignedTo	M:N (Appointment ↔ HealthcareProfessional)
BelongsTo	M:1 (HealthcareProfessional → Department)
Has	1:1 (Patient → MedicalRecord)
Has	1:M (MedicalRecord → MedicalEntry)
Updates	M:N (MedicalEntry ↔ HealthcareProfessional)
Generates	1:1 (Appointment → Bill)
Has	1:M (Bill → InsuranceClaim)
Prescribes	M:N (HealthcareProfessional ↔ Medication via Prescription)
ManagesInventory	M:N (Pharmacy ↔ Medication)
OrdersTest	1:M (HealthcareProfessional → LabTest)
HasResults	1:M (LabTest → LabResult)
InvolvedIn	1:M (Patient → EmergencyCase)
Treated	M:N (HealthcareProfessional ↔ EmergencyCase)

➤ Task#4 Assumption

- Appointments may involve group sessions; hence, Patient ↔ Appointment is M:N.
- Medical Record is a parent entity to Medical Entries that get updated each visit.
- Emergency cases are separate and have their own treatment and billing path.
- Medications prescribed are managed via a Prescription entity linking to both Patient and Doctor.
- Bills can exist without insurance claims, and vice versa, for private payers.
- Lab Tests and their results are tightly linked to appointments and Medical Records.

➤ Task#5 ER Diagram



Question# 4:

The administrators of Punjab University want you to design database for their motor vehicle pool. Given that brief summary of operations, draw the appropriate (and fully labelled) ERD. A brief description of operations is as follows:

- Punjab University has many departments.
- Punjab University has many faculty members.
- **A faculty member**, a student can register in **one department at a time**.
- Faculty members may use the vehicles owned by Punjab University for officially sanctioned travel. For example, the vehicles may be used by faculty members to travel to off-campus learning centers, to travel to locations at which research papers are presented, to transport students to officially sanctioned locations, and to travel for public service purposes. The

vehicles used for such purposes are managed by Punjab University's Travel Far But Slowly (TFBS) Centre.

- Using reservation forms, each department can reserve **vehicles** for its faculty, who are responsible for filling out the appropriate trip completion form at the end of a trip. **The reservation form** includes the expected departure date, vehicle type required, destination, and name of the authorized faculty member. The faculty member arriving to pick up a vehicle must sign a checkout form to log out the vehicle and pick up a trip completion form. (The TFBS employee who releases the vehicle for use also signs the checkout form.) The faculty member's **trip completion form** includes the faculty member's identification code, the vehicle's identification, the odometer readings at the start and end of the trip, maintenance complaints (if any), gallons of fuel purchased (if any), and the Punjab University credit card number used to pay for the fuel. If fuel is purchased, the credit card receipt must be stapled to the trip completion form. Upon receipt of the faculty trip completion form, the faculty member's department is billed at a mileage rate based on the vehicle type (sedan, station wagon, panel truck, minivan, or minibus) used. (*Hint: Do not use more entities than are necessary. Remember the difference between attributes and entities!*)
- All vehicle maintenance is performed by TFBS. Each time a vehicle requires maintenance, a maintenance log entry is completed on a **pre-numbered maintenance log form**. The maintenance log form includes the vehicle identification, a brief description of the type of maintenance required, the initial log entry date, the date on which the maintenance was completed, and the identification of the mechanic who released the vehicle back into service. (Only mechanics who have an inspection authorization may release the vehicle back into service.)
- As soon as the log form has been initiated, the log form's number is transferred to a **maintenance detail form**; the log form's number is also forwarded to the parts **department manager**, who fills out a **parts usage form** on which the maintenance log number is recorded. The maintenance detail form contains separate lines for each maintenance item performed, for the parts used, and for identification of the mechanic who performed the maintenance item. When all maintenance items have been completed, the maintenance detail form is stapled to the maintenance log form, the maintenance log form's completion date is filled out, and the **mechanic** who releases the vehicle back into service signs the form. The stapled forms are then filed, to be used later as the source for various maintenance reports.
- TFBS maintains a parts inventory, including oil, oil filters, air filters, and belts of various types. The parts inventory is checked daily to monitor parts usage and to reorder parts that reach the "minimum quantity on hand" level. To track parts usage, the parts manager requires each mechanic to sign out the parts that are used to perform each vehicle's maintenance; the parts manager records the maintenance log number under which the part is used.
- Each month TFBS issues a set of **reports**. The reports include the mileage driven by vehicle, by department, and by faculty members within a department. In addition, various revenue reports are generated by vehicle and department. A detailed parts usage report is also filed each month. Finally, a vehicle maintenance summary is created each month.

➤ Answer#4:

➤ Task#1: Entity Classes & Attributes

1. Department

- Attributes: Department-ID, Department-Name

2. **FacultyMember**
 - Attributes: Faculty-ID, Faculty-Name, Department-ID
3. **Vehicle**
 - Attributes: Vehicle-ID, Vehicle-Type, Registration-No
4. **ReservationForm**
 - Attributes: Reservation-ID, Expected-Departure-Date, Requested-Vehicle-Type, Destination, Faculty-ID
5. **CheckoutForm**
 - Attributes: Checkout-ID, Reservation-ID, Checkout-DateTime, TFBS-Employee-ID
6. **TripCompletionForm**
 - Attributes: TripCompletion-ID, Checkout-ID, Faculty-ID, Vehicle-ID, Odometer-Reading-Start and End, Fuel-Purchased, CreditCard-No, Receipt-Attachment, Billing-Amount
7. **MaintenanceLog**
 - Attributes: ML-ID, Vehicle-ID, MDetails, MSDate, MCDate, Mechanic-ID
8. **MaintenanceDetail**
 - Attributes: Detail-ID, Log-ID, Maintenance-Item, Part-ID, Mechanic-ID
9. **Part**
 - Attributes: Part-ID, Part-Name, Quantity
10. **PartsUsage**
 - Attributes: Usage-ID, ML-ID, Part-ID, Quantity-Used
11. **TFBS_Employee**
 - Attributes: Employee-ID, Employee-Name, Role

➤ Task#2: Relationships

1. **Registers**
 - A Department registers many FacultyMembers
→ 1:M from Department to FacultyMember
2. **Submits**
 - A FacultyMember makes vehicle reservations by submitting ReservationForms
→ 1:M from FacultyMember to ReservationForm
3. **Processes**
 - Each ReservationForm is processed to produce one CheckoutForm
→ 1:1 from ReservationForm to CheckoutForm
4. **CheckoutForm_TripCompletionForm (C_T)**
 - Each CheckoutForm is followed by one TripCompletionForm
→ 1:1 from CheckoutForm to TripCompletionForm
5. **Uses**
 - Each CheckoutForm is associated with one Vehicle, and a Vehicle may be used in many checkouts
→ 1:M from Vehicle to CheckoutForm
6. **Undergoes**
 - Each Vehicle undergoes maintenance recorded in MaintenanceLogs
→ 1:M from Vehicle to MaintenanceLog
7. **Contains**
 - Each MaintenanceLog contains one or more MaintenanceDetail entries

→ 1:M from MaintenanceLog to MaintenanceDetail
(Note: MaintenanceDetail is a weak entity identified by its parent MaintenanceLog.)

8. **SignsOut**

- Each CheckoutForm is signed out by one TFBS_Employee, and one TFBS_Employee may sign many CheckoutForms
- 1:M from TFBS_Employee to CheckoutForm

9. **Releases**

- Each MaintenanceLog is released (i.e., approved upon completion) by one TFBS_Employee (Mechanic)
- 1:M from TFBS_Employee to MaintenanceLog

10. **Records**

- Each MaintenanceLog may record multiple PartsUsage entries
- 1:M from MaintenanceLog to PartsUsage

11. **Supplies**

- Each PartsUsage entry refers to one Part; a Part can appear in many PartsUsage records
- 1:M from Part to PartsUsage

➤ **Task#3: Connectivity Summary**

Relationship

Connectivity

Registers	1:M (Department → FacultyMember)
Submits	1:M (FacultyMember → ReservationForm)
Processes	1:1 (ReservationForm → CheckoutForm)
C_T	1:1 (CheckoutForm → TripCompletionForm)
Uses	1:M (Vehicle → CheckoutForm)
Undergoes	1:M (Vehicle → MaintenanceLog)
Contains	1:M (MaintenanceLog → MaintenanceDetail)
SignsOut	1:M (TFBS_Employee → CheckoutForm)
Releases	1:M (TFBS_Employee → MaintenanceLog)
Records	1:M (MaintenanceLog → PartsUsage)
Supplies	1:M (Part → PartsUsage)

➤ **Task#4 Assumption**

- Appointments (i.e., vehicle reservations, checkouts, and trip completions) are recorded as separate entities to maintain a clear process flow.
- Faculty members are assumed to be the only users making vehicle reservations, even though other groups (such as students) might register in the same department.
- Each CheckoutForm is tied to a single Vehicle, but a Vehicle may be used repeatedly over time.
- MaintenanceDetail is modeled as a weak entity whose complete key is formed by combining Log-ID with Detail-ID.
- Parts usage during vehicle maintenance is tracked separately in the PartsUsage entity to support detailed reporting.

- TFBS_Employee records cover various roles (e.g., checkout clerk, mechanic, parts manager) and are differentiated by a Role attribute.

➤ Task#5 ER Diagram

